



Effect of particle size on the thermal and mechanical properties of aluminum composites reinforced with SiC and diamond

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ABSTRACT

The effect of particle size on thermal conductivity (TC) and bending strength was investigated in the SiC/Al and diamond/Al (Dia/Al) composites for thermal management applications. The results showed that with the particle size of SiC increasing from 7 to 250 μm , the TC of the 40 vol.% SiC/Al composite was improved by 50%, from 182 to 272 W/mK, while the bending strength decreased by 100% from 352 to 175 MPa. Comparatively, with diamond particles increasing from 30 to 200 μm , the TC of the 40 vol.% Dia/Al composite increased by around 52% from 313 to 475 W/mK, and the bending strength declined by 32% from 327 to 223 MPa. Characterization of fracture surface indicated that large SiC particles featured by an irregular shape tended to fracture prior to the Al matrix during loading, while large diamond particles with a regular shape did not have such a tendency. Moreover, diamond particles with a high intrinsic TC contributed to global TC much more than SiC ones with a relatively low TC. Thus, in order to balance thermal and mechanical properties of particulate reinforced aluminum composites, particle size of reinforcements should be well compromised according to their intrinsic properties.

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1. Introduction

With increasing power density and minimization of electronic components, the need for developing efficient thermal management materials with a high thermal conductivity (TC) and a low coefficient of thermal expansion (CTE) has become urgent [1–3]. To meet this requirement, aluminum matrix composites reinforced with SiC and/or diamond particles (hereafter SiC/Al and Dia/Al composites for short, respectively) have been developed in the past decades [3–9]. SiC/Al composite, with the TC typically in the range of 180–270 W/mK [8,10,11], is popularly used nowadays due mainly to its low cost and low CTE being compatible with that of electronic components, while Dia/Al composite with the much higher TC of around 350–750 W/mK [5–7,12–15], is the potential candidate for a niche market where specific products with an extremely high TC are required.

In general, global TCs of both SiC/Al and Dia/Al composites are determined by intrinsic TCs of SiC (244–400 W/mK for 6H-SiC [5,8,10,16–19]), diamond particles (1200–2000 W/mK [4,6,13,15]) and pure Al matrix (around 220–237 W/mK [5,13]) as well as the SiC/Al and Dia/Al interfaces. The resistance of thermal exchange across an interface can be described by interfacial thermal resistance (ITR). It has been reported that ITR can be significantly tailored by processing conditions

[6,20], surface coating of reinforcement particles [12,14,21–23] and matrix alloying [24,25]. Although the effect of chemical nature of interfacial bonding and the bonding strength on ITR remains unclear [10,15,26], the common understanding is that at the given volume fraction of particles, the use of fine particles introduces large interface area, which increases the total thermal resistance and, in turn results in a low global TC. For example, it has been reported that the TC of the liquid-infiltrated 65 vol.% Dia/Al composite increased from 400 to 760 W/mK by increasing diamond particle size from 30 to 150 μm [27]. In addition, efforts have been devoted to increasing the volume fraction of particles to enhance the global TC. For example, in the liquid-infiltrated SiC/Al composites, the content of SiC particles was as high as 74 vol.% via a bimodal particle size distribution, which increased the TC from 193 to 228 W/mK [9]. Hence, both increases of size and volume fraction of reinforcement particles are two practical means to improve global TC of these composites.

In addition to thermal properties, mechanical properties of such composites, e.g. bending strength, is also rigidly required for thermal management applications according to their functions of mechanically supporting protection and heat dissipation for electronic components [28–30]. As previously reported [31], the strengthening mechanisms of particulate reinforced metal matrix composites (PRMMCs) are two-fold. Firstly, load can be effectively transferred from the plastic matrix to the hard and rigid particles, while large particles are prone to fracture themselves during deformation, thus the pre-existing cracks initiate

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fractures and cannot ensure load transfer. For example, the yield strength and plastic work hardening rate of the SiC/Al composites would decrease with increasing the particle size [31]. Secondly, the metal matrix surrounding the particles can be strengthened by increasing dislocation density due to thermal mismatch during cooling. But the effect of thermal mismatch strengthening would be weakened with reduced interface area by using large particles. It should be noted that the content of rigid particles in most of PRMMCs for structural applications is usually below 40 vol.%, since a higher content seriously degrades the deforming ability. Contrary to thermal properties of the PRMMCs, it's not beneficial for mechanical properties by using large particles with a high content. Hence, a compromised solution should be proposed to design the SiC/Al and Dia/Al composites for thermal management applications in order to have high TC values without sacrificing necessary bending strength.

In our previous work, the SiC/Al and Dia/Al composites, with TCs as high as around 270 and 570 W/mK [10,13] respectively, have been produced by vacuum hot pressing (VHP) under the optimized conditions. This work aims to further study the effect of particle size variation on the TC and bending strength of SiC/Al and Dia/Al composites. It also provides an example demonstrating that, the significance of simply using large reinforcement particles should be conscientiously reconsidered in the design of PRMMCs for thermal management applications.

2. Experimental procedure

Atomized aluminum powders with a mean size of 10 μm and 99.9% in purity were used as the matrix material. Green α -SiC particles with mean sizes of 7, 40, 70, 160 and 250 μm and 99.4% in purity, and synthetic diamond particles (type HWD40) of 30, 75 and 200 μm , were used as the reinforcements. As shown in Fig. 1, typically the SiC and fine diamond particles showed irregular shapes, while the large diamond particles (200 μm in size) showed a tetrakaidecahedron regular shape with the {111} and {100} faces clearly identifiable. The reinforcement particles were firstly blended with the Al powders and cold-pressed at room temperature. The compact was then sintered in a

graphite mold by VHP. The sintering temperature was set to 650 $^{\circ}\text{C}$. The holding time was set to 1 h for SiC/Al and 2 h for Dia/Al composites. The volume fraction of the particles were 40 vol.% in both composites. Detailed information of the VHP process can be found in [10,13].

Relative density was measured by the Archimedes method. The TC (λ) of the sintered composites at room temperature was calculated from thermal diffusivity (α), density (ρ) and specific heat capacity (C_p) in terms of the equation: $\lambda = \alpha \times c_p \times \rho$. Thermal diffusivity was measured by the laser-flash technique using a Netzsch LFA447 thermal constant analyzer. Each thermal diffusivity value was an average of three measurements and the standard deviation is 2%. Specific heat capacity was calculated by the rule of mixtures (ROM) using the specific heat capacity of Al (0.895 J/g K), SiC (0.66 J/g K) and diamond (0.45 J/g K). Three-point bending test was performed at ambient temperature on a Zwick universal test machine with specimen of 10 $\times$ 10 $\times$ 40 mm in size. Morphologies of SiC and diamond particles, and fracture surface of the sintered composites were characterized by scanning electron microscopy (SEM) with an FEI Quanta FEG 250 instrument.

3. Results

3.1. Effect of particle size on thermal conductivity

Fig. 2 shows the relative density of the sintered SiC/Al and Dia/Al composites. The relative density first increased to 98.9% with particle size increasing from 7 to 70 μm for the SiC/Al composites, and then slightly decreased to 98.5% with further increasing particle size. Comparatively, the relative density of the Dia/Al composites declined from 98.6 to 97.1% with particle size increasing from 30 to 200 μm , probably due to practical difficulty in consolidation of the composites with large particles.

Fig. 3 shows the global TC and bending strength of the SiC/Al and Dia/Al composites. For the SiC/Al composites, the TC increased by around 41% from 182 to 256 W/mK when the particle size increased from 7 to 40 μm . There is not clear TC variation with particle size

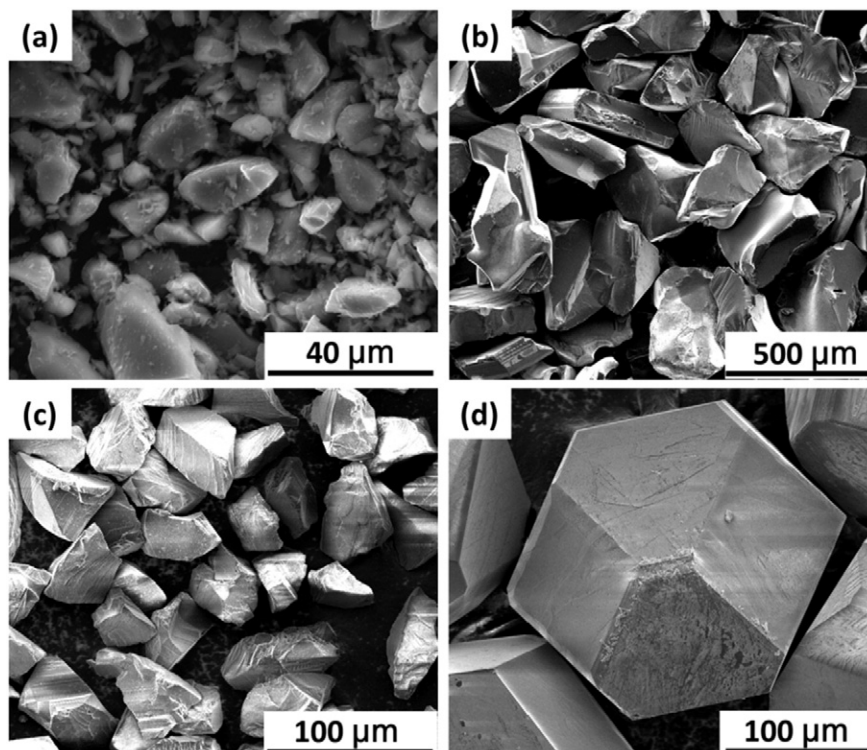


Fig. 1. SEM images showing typical morphologies of SiC particles with mean sizes of (a) 7 and (b) 250 μm , and diamond particles with mean sizes of (c) 30 and (d) 200 μm .

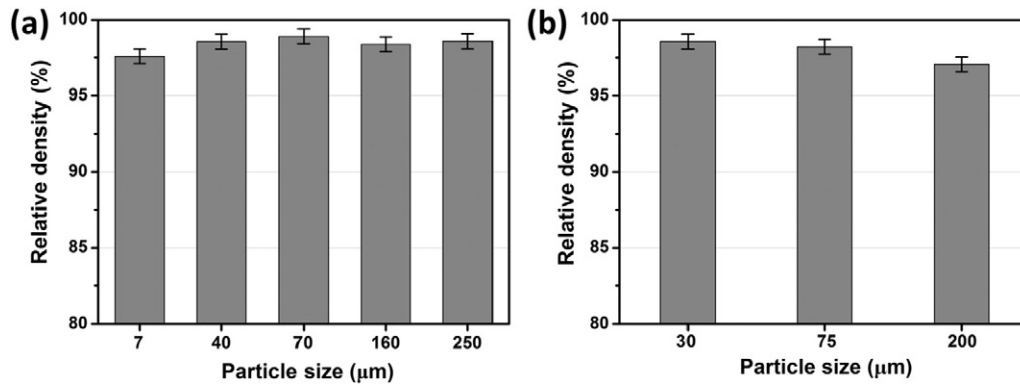


Fig. 2. Relative densities as a function of particle size in (a) the SiC/Al and (b) Dia/Al composites.

increasing in the range of 40–160 μm, and the highest TC was 272 W/mK with the largest particle size of 250 μm. However, for the Dia/Al composites, the TC showed a nearly linear increase by 51%, from 313 to 475 W/mK, with particle size increasing from 30 to 200 μm.

3.2. Effect of particle size on bending strength and fractography

Bending strength of the SiC/Al and Dia/Al composites was shown in Fig. 3. For both composites, the bending strength decreased monotonically with increasing particle size. For the SiC/Al composites, the highest bending strength of around 350 MPa was obtained with the particle size of 7 μm, while the lowest one was around 175 MPa with the particle size of 250 μm, indicating a decline of 50%. For the Dia/Al composites, the highest bending strength of 327 MPa was obtained when the particle size was 30 μm, being close to the highest one in the SiC/Al composites. With the particle size increasing to 200 μm, the bending strength of the Dia/Al composites decreased by around 32% to 223 MPa. Thus, a sharp decrease by 50% occurred in the SiC/Al composites when the particle size increased from 40 to 160 μm, while no such a feature was observed in the Dia/Al composites.

Fracture surfaces of the SiC/Al and Dia/Al composites after bending test were shown in Figs. 4 and 5. As shown in Fig. 4, neither decohesion of the SiC/Al interface nor pores in the Al matrix are visible in the SiC/Al specimens. This indicates a strong interfacial bonding due to sufficient consolidation. In the SiC/Al composites with the particle sizes of 7 and 40 μm, plastic dimples in the Al matrix can be clearly observed, while fracture in the SiC particles is invisible. With the particle size larger than 70 μm, though some dimples can still be seen, some fractures in SiC particles appear, as shown by arrows in Fig. 4f, h and j.

Furthermore, the fracture surface of the Dia/Al composite reinforced with diamond particles of 30 μm (Fig. 5a) shows that the particles are tightly embedded in the Al matrix, and no interfacial fracture is visible,

indicating the strong interfacial bonding. Dimples can be seen to be located between the diamond particles, which indicates plastic deformation of the Al matrix and efficient load transfer. This phenomenon is similar to that of the SiC/Al composites with the particle sizes of 7 and 40 μm (see Fig. 4). However, some fine diamond particles are fractured, as shown by arrows in Fig. 5a. For the Dia/Al composites reinforced with particles of 200 μm (Fig. 5b), diamond particles are tightly embedded in the Al matrix, and dimples and residual Al on the {100} faces of diamond particles can also be seen, a sign of good interfacial bonding and even stronger than the strength of Al matrix, due to the different carbon diffusion abilities on these faces [13,26]. However, the {111} faces are still bare without effective bonding with the Al matrix, which is unfavorable to minimize the ITR. The preferential interfacial bonding is in accordance with the previous reports [13,14,26].

4. Discussions

4.1. Effect of particle size on thermal conductivity

As revealed above by experimental evidences, the effects of particle size variation on TC and bending strength of the SiC/Al and Dia/Al composites are quite different. In order to understand the different responses, the TC of the SiC/Al composites is theoretically analyzed by Hasselman–Johnson Model [17]:

$$K_c = \frac{K_m [2K_m + K_r^{eff} + 2(K_r^{eff} - K_m)V_r]}{2K_m + K_r^{eff} - (K_r^{eff} - K_m)V_r} \quad (1)$$

$$K_r^{eff} = \frac{1}{\frac{1}{K_r^{in}} + \frac{2}{Dh}} \quad (2)$$

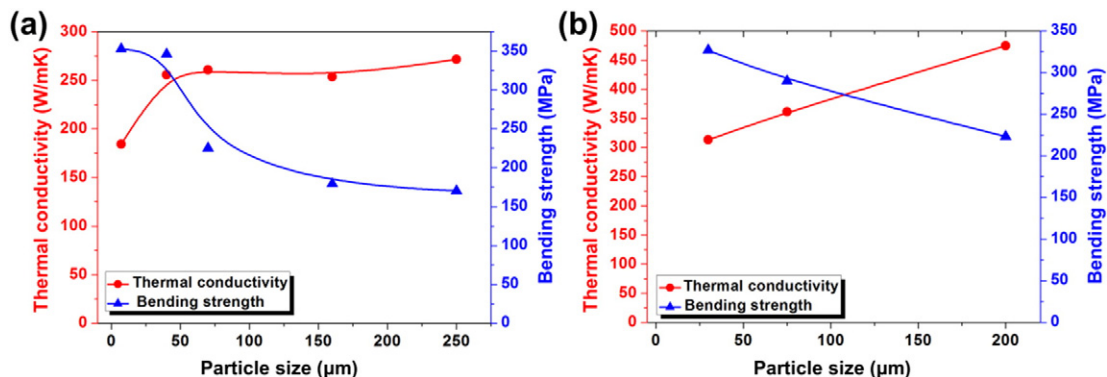


Fig. 3. Global TC and bending strength of (a) the SiC/Al and (b) Dia/Al composites with different particle size.

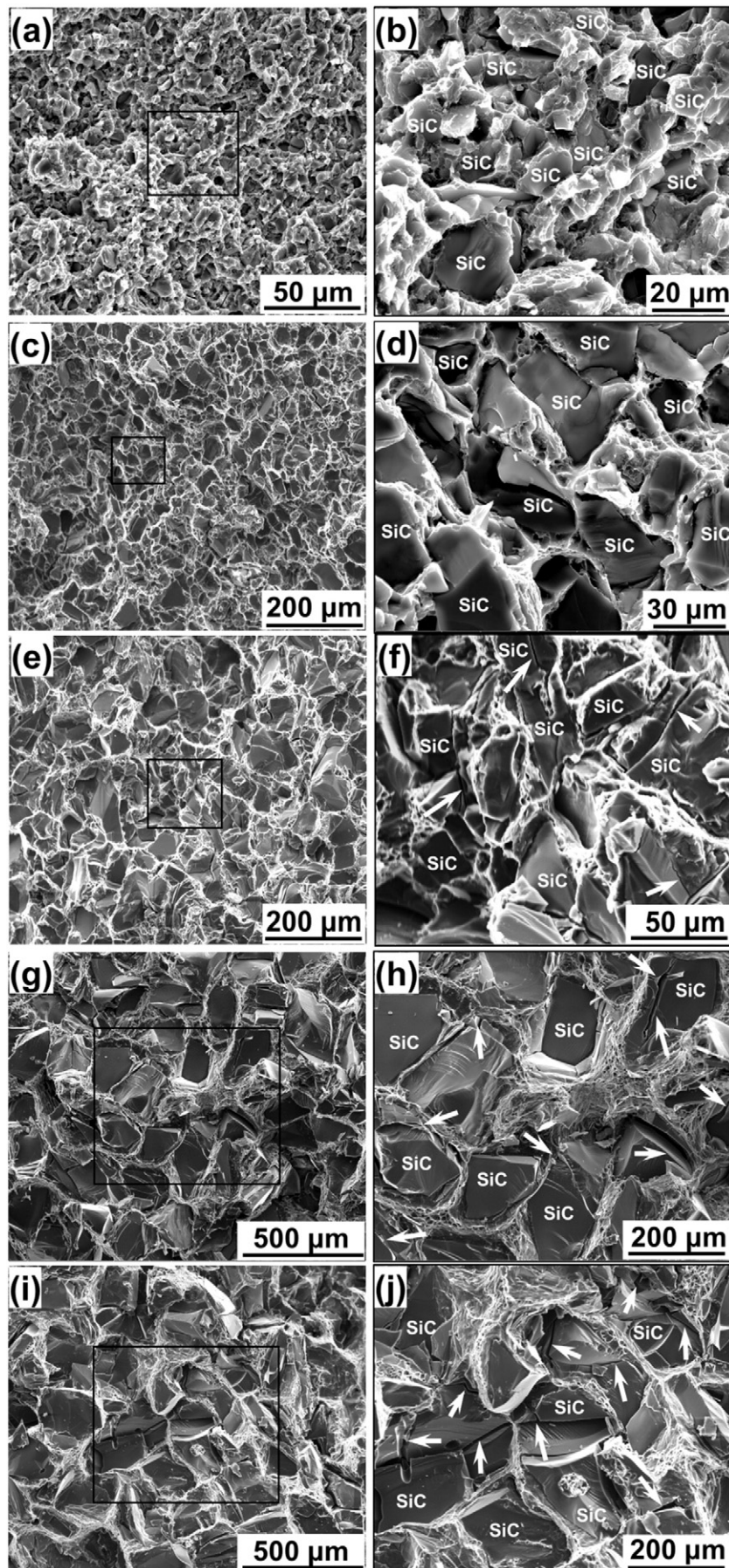


Fig. 4. Fracture surface of the SiC/Al composites with different particle sizes: (a) (b) 7, (c) (d) 40, (e) (f) 70, (g) (h) 160 and (i) (j) 250 μm .

where subscripts c , m and r stand for the composite, matrix and reinforcement, respectively. K , V , D and h are the TC, volume fraction average diameter of particles, and interfacial thermal conductance between reinforcement and matrix, respectively. The superscripts eff and in denote the effective and intrinsic TC of a reinforcement. As pores are supposed to mainly exist in the matrix and can be treated as non-thermally conductive inclusions, the effective TC of Al matrix can be expressed as [13]:

$$K_m = \frac{K_m^{in}(1 - V_p)}{1 + \frac{1}{2}V_p} \quad (3)$$

where the subscript p stands for pores. K_m^{in} is taken as 233 W/mK.

To get more precise approach, the TC of the Dia/Al composites with high phase contrast was analyzed by Differential Effective Medium scheme [13], and the TC was expressed as:

$$(1 - V_r) \left(\frac{K_c}{K_m} \right)^{\frac{1}{2}} = \frac{K_r^{eff} - K_c}{K_r^{eff} - K_m} \quad (4)$$

To understand the effect of D and K_r^{in} on the TC of the composites, K_r^{eff} values of the SiC and diamond particles were back-calculated from the measured TC and the measured porosity in the composites following the way as described in ref. [13].

It can be seen from Fig. 6(a) and (b) that K_r^{eff} and K_c show a similar variation in both SiC/Al and Dia/Al composites. For the SiC/Al composites, both K_r^{eff} and K_c show a rapid increase with particle size smaller than 70 μm , followed by a plateau, with the effective TC finally approaching the intrinsic values (375 W/mK). For the Dia/Al composites, however, both K_r^{eff} and K_c increase greatly with particle size increasing, and an overall linear relation with particle size can be seen. Even for the composite with particle size of 200 μm , the value of K_r^{eff} is still much smaller than the intrinsic value of diamond (1750–1800 W/mK [13]), which indicates there is still a large margin for further enhancing K_c .

Except for the additional ITR caused by the interfacial debonding, as shown in Figs. 4 and 5, the different responses of K_r^{eff} to particle size, D , can be explained by the difference of the intrinsic TC of diamond (1200–2000 W/mK [4,6,13,15]) and SiC particles (244–400 W/mK for 6H-SiC [5,8,10,16–19]), K_r^{in} , as shown in Eq. (2). With D increasing, K_r^{eff} will gradually approach K_r^{in} . Thus, when D reaches the critical value, there will be only a negligible increment of K_r^{eff} by further increasing particle size, and consequently no observable increase in global TC can be seen. As expected, for the 40 vol.% SiC/Al composites in this study, the critical value is around 70 μm , while for the 40 vol.% Dia/Al counterparts, this value seems to be even larger than 200 μm due to the high intrinsic TC and the additional ITR caused by the preferential interfacial bonding as discussed in Section 3.2. In this regard, the higher the intrinsic TC of the reinforcement is, the larger critical particle size is expected, and greater benefit by using large particles can be

expected. This may be the right reason that diamond particles with the size even larger than 400 μm were sometimes used for the production of high thermally conductive metal matrix composites [5–7,12,32–36], while SiC particles usually smaller than 100 μm [8,9,11,16,18,19,37,38].

4.2. Effect of particle size on bending strength and fractography

According to Fig. 3, for the SiC/Al composites, a sharp decrease in the bending strength (by around 50%) occurs when the particle size increases from 40 to 160 μm , while no such a feature can be observed in the Dia/Al composites. This difference could be explained by the dimples and SiC (or Dia) particles observed in the fracture surface (Figs. 4 and 5).

Dimples are the direct consequences of plastic deformation of the Al matrix. The presence of fractures in the SiC particles themselves indicates that the large particles tend to fracture prior to the Al matrix. This can lead to crack propagation and decrease load bearing efficiency from the metal matrix, as evidenced by the decreased number of dimples with the particle size increasing.

According to Fig. 5, in the Dia/Al composites reinforced with the particles of 30 and 200 μm in size, diamond particles were tightly embedded in Al matrix and no interfacial fracture but dimples and residual Al could be seen, which indicates a strong interfacial bonding and sufficient load transfer. As shown in Figs. 4 and 5, the particle fracture behaviors are probably mainly governed by their shape and intrinsic strength. The predominant causes for the decline of bending strength are different in the SiC/Al and Dia/Al composites. Due to the fracture of large SiC particles themselves, load transfer and matrix deformation are both seriously diminished in the SiC/Al composites reinforced with large particles, thus the bending strength drastically declines. However, the dimples in the matrix instead of particle fractures are clearly visible in the Dia/Al composites reinforced with large particles, thus their declined bending strength mainly results from the reduced interface area, which in turn weakens the strengthening of interfacial load transfer and thermal mismatch.

4.3. Effect of particle size on the comprehensive performances

To consider the comprehensive performance of both composites, the disadvantages of large SiC particles appeared to be obvious. As shown in Fig. 3a, the use of large SiC particles, e.g. with particle size larger than 70 μm , will lead to a major degradation in bending strength, but have a very limited effect on TC. The nonlinear changes in mechanical and thermal properties make the SiC particle size of 40 μm being the most suitable for comprehensive performance. In contrast, the use of large diamond particles will contribute to a linear TC enhancement, even with particle size larger than 200 μm , while maintain a moderate linear decreased bending strength.

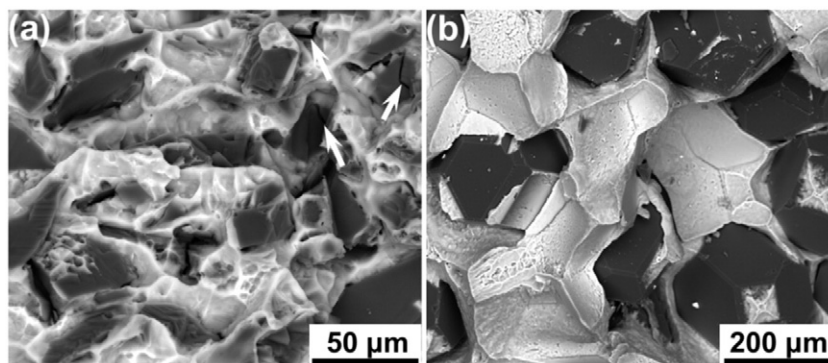


Fig. 5. Fracture surface of the Dia/Al composites with different particle sizes: (a) 30 and (b) 200 μm .

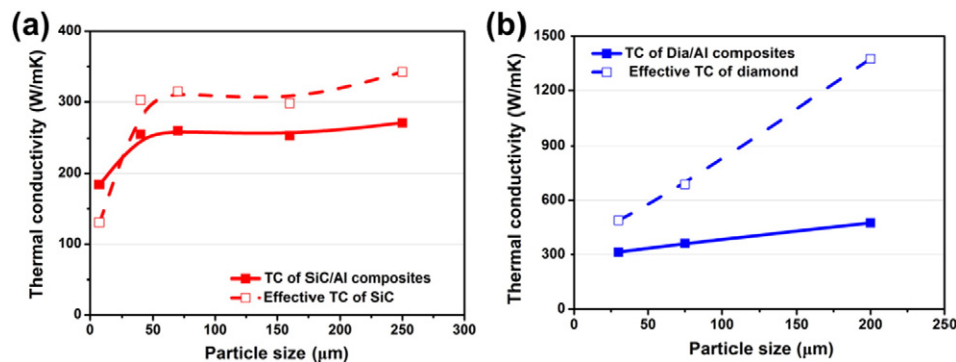


Fig. 6. Experimentally measured and calculated effective TCs of (a) the SiC/Al and (b) Dia/Al composites with different particle sizes.

It can be inferred that, with particle sizes increasing, the TC and bending strength of PRMMCs strongly depends on intrinsic characteristics of the incorporated particles. The higher intrinsic TC of the particles is, a higher critical particle size is expected, and a global TC increase can be achieved in a broader range of particle size. The bending strength of particulate reinforced composites decreases more drastically when the particles have lower intrinsic strength or irregular shape, a transition in failure mechanism, from the plastic fracture of matrix to the brittle fracture of particles, may even lead to a nonlinear decline. Therefore, particle size should be optimized to neither exceed the critical value inducing a marginal increment of K_p^{eff} , nor lead to large-scale transition in failure mechanism. As a result, the permitted upper limit of particle size is higher for those particles with regular shape and high strength, e.g. diamond, than those with irregular shape and lower strength, e.g. SiC. Thus, the real value by using large particles to improve thermal property, irrespective of the mechanical property of the composites, should be discreetly considered in the design of PRMMCs for thermal management applications.

5. Conclusions

The effect of particle size on TC and bending strength of the VHPed SiC/Al and Dia/Al composites was studied. The main conclusions can be drawn as follows:

- 1) Large SiC particles with a low intrinsic TC contributed marginally to the global TC of the SiC/Al composites, but seriously degraded the bending strength. With SiC particle size increasing from 7 to 250 μm , the TC of the SiC/Al composites increased from 182 to 272 W/mK, while the bending strength decreased from 353 to 170 MPa.
- 2) Large diamond particles with a high intrinsic TC greatly contributed to the global TC of the Dia/Al composites, and slightly degrade their bending strength. With diamond particle size increasing from 30 to 250 μm , the TC of the Dia/Al composites linearly increased from 313 to 475 W/mK, while the bending strength linearly decreased from 327 to 223 MPa.
- 3) The different responses of SiC and diamond particles to the thermal and mechanical properties of the composites are mainly attributed to their different intrinsic properties. Large SiC particles tend to fracture during deformation and thus deteriorate the bending strength of the composites, while large diamond particles do not. This study indicates that, in the design of thermal management materials, the significance of simply using large particles should be conscientiously considered.

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